Gas quadrangle — Solution

X20 (2020) — English translation

A1 — 0.80 pts

Let the pipe be adiabatically insulated. Obtain the dependence of gas velocity on its temperature.

Answer: Answer: ZSE

$$P_0 dV_0 - P dV = dm\left(\frac{v^2 - v_0^2}{2} + \frac{C_V}{\mu}(T - T_0)\right)$$

A2 — 0.70 pts

Show that when gas flows, the relation

$$v\frac{dv}{dx} = -\frac{1}{\rho}\frac{dP}{dx}.$$

is satisfied

The law of momentum change for gas

$$-Sdp = adm$$

Equality $a = v\frac{dv}{dx}$ proves the required

A3 — 0.50 pts

Now thermal power $N = kL$ is supplied to any section of pipe with length L. Derive an expression for the gas velocity as a function of x and T.

ZSE

$$Ndt = P_0 dV_0 - P dV = dm\left(\frac{v^2 - v_0^2}{2} + C_V(T - T_0)\right)$$

Conduct transformation,obtain

$$v^2 = v_0^2 + \frac{2C_P}{\mu}(T_0 - T) + \frac{2N}{m}$$

Answer

$$v = \sqrt{v_0^2 + \frac{2C_P}{\mu}(T_0 - T) + \frac{2kx}{\rho_0 S_0 v_0}}$$

A4 — 0.50 pts

Now the gas temperature depends linearly on x according to the law $T = T_0 + \alpha x$. Show that the gas moves through the pipe with uniform acceleration and find this acceleration a.

$a = v\frac{dv}{dx} = -\frac{\alpha C_P}{\mu} + \frac{k}{\rho_0 S_0 v_0} = const$, which proves the required

A5 **0.50 pts**

Obtain an equation for a polytropic process with constant molar heat capacity C in coordinates (P, ρ).

$$C = C_V + \frac{\nu RT}{V}\frac{dV}{dT}$$

$$After integration have$$

TV$^{n-1} = const$

$$P\rho^{-n} = const$$

$$where$$

n=C$_P - C\frac{}{C_V - C}$

A6 **1.20 pts**

Show that under the conditions of point A4 the gas moves as if it were in a polytropic process.

From the equation of state and condition $\alpha = \frac{dT}{dx} = const$ we have

$$a = v\frac{dv}{dx} = -\frac{1}{\rho}\frac{dP}{dx} = -\frac{RT}{\mu P}\frac{dP}{dT}\frac{dT}{dx} = -\frac{\alpha R}{\mu}\frac{T}{P}\frac{dP}{dT}$$

$$But since$$

a=-$\alpha C_P\frac{}{\mu + \frac{k}{\rho_0 S_0 v_0} = const}$

$\frac{T}{P}\frac{dP}{dT} = \frac{\mu}{\alpha R}\left(\frac{\alpha C_P}{\mu} - \frac{k}{\rho_0 S_0 v_0}\right) = const$10 After integration we get:

$$TP^{\frac{1}{\frac{\mu k}{\alpha R \rho_0 S_0 v_0} - \frac{C_P}{R}}} = TP^{\frac{R}{C - C_P}} = const$$

$$,$$

C=$\mu k\frac{}{\alpha \rho_0 S_0 v_0}$

A7 **0.80 pts**

Derive an expression for the gas density as a function of x.

From the polytropic equation

$$\rho = \rho_0\left(\frac{T}{T_0}\right)^{\frac{1}{1-n}}$$

$$Find$$

n$from$A6$, obtain \rho = \rho_0\left(\frac{T_0}{T_0 + \alpha x}\right)^{1 - \frac{C_P}{R} + \frac{\mu k}{\alpha R \rho_0 S_0 v_0}}$

B1 **0.10 pts**

Find the gas mass flow rate $\dot{m}$.

$$\dot{m} = \rho_0 S_0 v_0$$

B2 **0.80 pts**

Find the heat $\delta Q_{ab}, \delta Q_{bc}, \delta Q_{cd}, \delta Q_{da}$ received by a portion of gas with mass dm in these areas of the quadrilateral. Answers can be expressed in terms of $v_a, S_a, T_a, \rho_a, \mu, C_P, \alpha, A, \beta, B, L_1, L_2, L_3, L_4$ and dm.

From the first law of thermodynamics and point $A6 \delta Q_{\overline{dm=\frac{C}{\mu}(T-T_0)}} Substitute in last equation expression for hea$

$$Q_1 = \frac{\alpha dm}{\rho_0 S_0 v_0} L_1$$

,$Q_2 = \frac{\beta dm}{\rho_0 S_0 v_0} L_2$,$Q_3 = -\frac{\alpha dm}{\rho_0 S_0 v_0} L_3$,$Q_4 = -\frac{\beta dm}{\rho_0 S_0 v_0} L_4$

B3 **1.00 pts**

Get β and B via $v_a, S_a, T_a, \rho_a, \mu, C_P, \alpha, A, L_1, L_2, L_3, L_4$.

Since the state is stationary, the quadrilateral receives zero thermal power. From here

$$\alpha(L_1 - L_3) + \beta(L_2 - L_4) = 0$$

Then answer on first question: $\beta = \alpha \frac{L_1 - L_3}{L_4 - L_2}$ Find difference temperature in point c and a two method

$$T_c - T_a = AL_1 + BL_2 = AL_3 + BL_4$$

Hence answer on second question:$B = A \frac{L_1 - L_3}{L_4 - L_2}$

B4 **1.50 pts**

Prove that for such a system to be stationary, it is necessary that the molar heat capacity of the gas be constant throughout the entire cycle.

At point $A6$ we received

$$C = \frac{\mu}{\rho_0 S_0 v_0} \frac{dN}{dx} \frac{dx}{dT}$$

$$For$$

abandcd

11

For bc and da

$$C_2 = \frac{\mu \beta}{B \rho_0 S_0 v_0}$$

Since из пункта B_3 следует,что $\frac{\alpha}{A} = \frac{\beta}{B}$

$$C_1 = C_2$$

Since the processes in all pipes are polytropic, the molar heat capacity of the gas is the same throughout the process

B5 **0.70 pts**

Find the accelerations a_{ab} and a_{bc} through $v_a, S_a, T_a, \rho_a, \mu, C_P, \alpha, A, L_1, L_2, L_3, L_4$.

Substituting A,B,α,β into the expression in point $A4$, we have

$$a_{ab} = -\frac{AC_P}{\mu} + \frac{\alpha}{\rho_0 S_0 v_0}$$

a$_{bc} = \left(-\frac{\alpha C_P}{\mu} + \frac{k}{\rho_0 S_0 v_0}\right) \frac{L_1 - L_3}{L_4 - L_2} = a_{ab} \frac{B}{A}$

B6 **0.50 pts**

Find the velocities v_b, v_c *and* v_d *through* $v_a, S_a, T_a, \rho_a, \mu, C_P, \alpha, A, L_1, L_2, L_3, L_4$.

Substituting the previously obtained results into the formula $v^2 = v_0^2 + 2a_x x$, we have

$$v_b^2 = v_0^2 + 2a_{ab}L_1$$

v$_c^2 = v_0^2 + 2(a_{ab}L_1 + a_{bc}L_2)$
v$_d^2 = v_0^2 + 2a_{bc}L_4$

B7 **0.20 pts**

Find the minimum time t *after which the gas returns to its initial position. Express your answer in terms of the velocities at the points of intersection of the pipes and the accelerations in the pipes (the item is not assessed unless B5 and B6 are completed).*

Let's express time through changes in speed in individual sections of the pipe and acceleration in them

$$t = \frac{v_b + v_c - v_a - v_d}{a_{ab}} + \frac{v_c + v_d - v_a - v_b}{a_{bc}}$$

B8 **0.20 pts**

Find the mass of gas in the pipe. Express your answer in terms of $\dot{m}$ *and* t *(the item is not scored unless B5 and B6 are completed).*

Mass of gas in the pipe $m = \dot{m}t$